

Knowledge Channels for LLM Agents on Remote-Sensing Tasks: A Factorial Study Across Model Scale and Task [Experiment]

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Abstract

LLM agents are increasingly used in remote-sensing analysis on optical satellite imagery, yet practitioners lack guidance on which knowledge channel actually helps. We benchmark seven open-weight backbones from 8 B dense to 1.6 T mixture-of-experts on five pixel-scored Sentinel-2 RS tasks, the fifth a new compositional spatial query combining class detection with distance-buffer construction. We vary three knowledge channels. KB is a hybrid-retrieval knowledge base, WS a live web-search tool, and SK a small library of task-specific Agent Skills SOPs.

Across the 280 model-configuration-task cells, F1 varies most across tasks at $\sigma=0.21$, ahead of model scale at $\sigma=0.08$ and knowledge channel at $\sigma=0.03$. At 27 B total parameters and above, no knowledge channel shifts panel-pooled F1 by more than 0.04 across five models from three providers, defining the Saturated regime. Two single-backbone observations lie outside this regime, a Decisive +0.274 F1 SK lift at 8 B and a Transition +0.060 F1 lift at 14 B, both awaiting replication on similar-scale backbones. Only 45 % of these cells exceed a published spectral-index baseline. SK is both the strongest score lever and the most efficient channel, raising the panel-pooled score by 0.060 F1, stable at 0.061 on perception tasks alone, while cutting panel-median tool calls from 13 to 8 and output tokens by about 40 %. Manual failure labelling attributes 91 % of failures to RS-specific choices of band, threshold, or class against 1.1 % to code execution, explaining why procedural skills dominate open-ended retrieval. On the new spatial query the residual bottleneck is upstream spectral class identification, not the distance-buffer operator. We release per-trial logs, the KB corpus, the skill documents, the spatial-query grammar, and failure-mode labels to support reproduction.

CCS Concepts

• **Computing methodologies** → **Natural language processing; Intelligent agents**; • **Information systems** → **Geographic information systems; Information retrieval**; • **General and reference** → **Empirical studies**.

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SIGSPATIAL '26, Riverside, CA, USA
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ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
<https://doi.org/10.1145/nnnnnnn.nnnnnnn>

Keywords

remote sensing, LLM agents, knowledge channels, procedural skills, retrieval-augmented generation, spatial query, factorial benchmark, model scaling

ACM Reference Format:

Yingjun Bian, Fan Yin, and Gang Wu. 2026. Knowledge Channels for LLM Agents on Remote-Sensing Tasks: A Factorial Study Across Model Scale and Task [Experiment]. In *The 34th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL '26)*, November 3–6, 2026, Riverside, CA, USA. ACM, New York, NY, USA, 13 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 Introduction

Recent remote-sensing automation increasingly relies on LLM agents. RS-Agent [50], ThinkGeo [40], OpenEarth-Agent [59], LLM-Find [36], and GIS-Copilot [1] all have the LLM emit Python code or tool calls that produce a geo-referenced raster. A recent survey [2] catalogues more than a dozen further single-task or single-backbone systems. Two practical questions for designers remain open: which combination of knowledge channels should an RS agent expose, and does the optimal choice change with backbone scale? We answer both with RS-AgentBench, a pixel-scored benchmark crossing three knowledge channels, seven open-weight backbones, and five RS tasks.

What is underspecified. Existing systems bundle their knowledge channels and rarely vary the backbone [2]. Which channel drives the gain, how the gain scales with model capacity, and whether the answer differs across task types are all unresolved. Studies that report ablations [50, 55] either bundle channels or fix the backbone. The most recent integrated RS agent [46] always deploys retrieval, web search, and perception together, leaving marginal contributions unknown. Scale-varying skill studies [28] operate within frontier models only and outside the RS domain.

A controlled framing. The knowledge-channel interface exposed to an LLM agent on RS tasks has three orthogonal axes: (a) a generic knowledge base (KB) retrieved via vector search; (b) live web search (WS) used as a tool; and (c) task-specific procedural skill SOPs (SK) that the agent autonomously loads on demand, following the Agent Skills paradigm [49]. Crossed with model scale and task type, the design becomes a $2^3 \times 7 \times 5$ factorial. We run it over five RS tasks (T1 water extraction, T2 change detection, T3 four-class land cover, T4 burn-scar detection, and the new T5 compositional spatial query), eight KB $\times$ WS $\times$ SK configurations, and seven open-weight backbones spanning 8 B dense to 1.6 T mixture-of-experts. We restrict to open-weight backbones for reproducibility and provider-agnostic

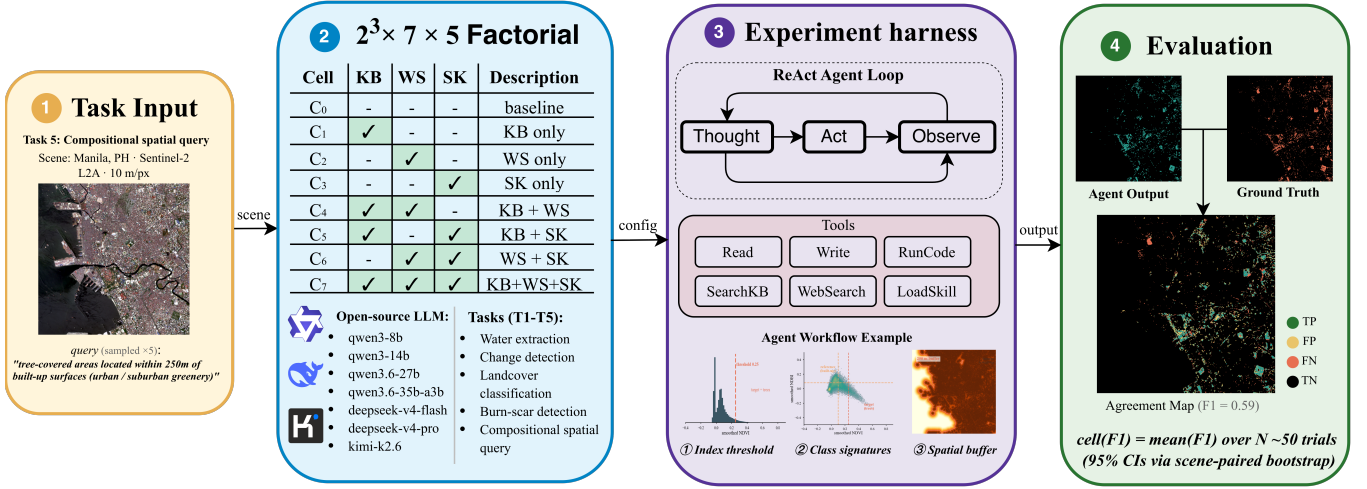


Figure 1: RS-AgentBench experimental architecture. Task input (raster + query) feeds an LLM agent under a 2^3 knowledge-channel factorial (KB / WS / SK), crossed with seven backbones and five RS tasks, scored against public ground-truth rasters.

token-cost metrics; closed-source families such as GPT, Claude, and Gemini are deferred to future work. Of the 280 design cells—one (channel-configuration, backbone, task) combination—268 receive $N \geq 50$ trials and the remaining 12 qwen3.6-35b-a3b cells are audited in §4.4; the matrix totals 14,002 scored trials. Full trial protocol and sampling settings are in §4.

Findings. The two questions admit a joint answer (Fig. 3, §5.1).

(1) Among the three channels, SK is the dominant lever: it raises panel-pooled F1 by +0.060 F1 while cutting panel-median tool calls from 13 to 8 ($\sim 38\%$ bare-to-SK, $\sim 50\%$ pooled) and output tokens by $\sim 40\%$, so it is both the strongest score channel and the most efficient. (2) The optimal choice attenuates with scale. Five ≥ 27 B backbones across three providers form a Saturated band at panel-pooled SK of +0.001 to +0.035 F1; outside this band, a single-backbone Decisive observation at qwen3-8b reaches +0.274 F1 with partially deterministic SK short-circuit, and a qwen3-14b-specific Transition anomaly reaches panel-pooled SK = +0.060 F1 where KB stacks additively on SK at +0.052 F1 via Skill-invocation re-weighting (§5.4). Both single-backbone observations await replication on non-Alibaba backbones at similar scale. The compositional spatial-query task is bounded by class identification from spectral indices, not by the spatial-buffer operator.

8B-specific patterns. The 8 B backbone shows two further patterns (§5.4): deterministic SK short-circuit—identical predicted masks across repetitions—and a net-harmful {KB, WS} combination at C₄, -0.038 F1 vs. C₀ on the five-task panel mean and -0.061 F1 on the four tasks T1–T4. The latter agrees with GeoAnalystBench’s DeepSeek-R1-7B finding [55] and disappears once SK is co-present.

Contributions.

- (C1) A new task for the suite: a compositional spatial query (T5) that decomposes deterministically into target-class detection, reference-class detection, distance-buffer construction (a classical spatial-join primitive [7, 21]), and mask intersection. The construction separates spectral class-grounding

errors from spatial-operator errors at sub-step level, derives ground truth deterministically from WorldCover, and pairs the task with four hand-written reference pipelines that establish a published spectral upper bound near 0.19 F1 against a label-oracle ceiling of 1.00. We instantiate the canonical *within* operator over six WorldCover target classes; other operators (*contains*, *nearby*, *intersects*) are deferred to follow-on work.

- (C2) The RS-AgentBench harness: a reproducible pixel-level raster scorer that evaluates spatial-analysis agents against public ground-truth rasters across five canonical RS task families on globally distributed Sentinel-2 scenes, under a 2^3 knowledge factorial (KB / WS / SK) and a seven-backbone scale gradient from 8 B dense to 1.6 T MoE. Unlike prior RS-agent benchmarks [40, 50, 55] that score via tool-call success or LLM-judge ratings, our harness uses the same pixel-mask metric against which RS systems are deployed. The resulting 14,002 scored trials, harness code, per-trial logs, KB corpus, SOPs, T5 grammar, and failure-mode labels are released for reproduction.
- (C3) A backbone-scale selection criterion for RS-agent knowledge-channel design, grounded in the factorial evidence. The primary, multi-model finding is the Saturated regime above ~ 27 B total parameters: five models spanning three providers with panel-pooled SK $\leq +0.04$ F1 and panel-mean $|\Delta_{C_5-C_3}| \leq 0.010$ F1, across which the choice of knowledge channel is largely irrelevant. The 8 B end-point qwen3-8b shows a single-model Decisive observation at panel-pooled SK = +0.274 F1 with partially deterministic SK short-circuit, and qwen3-14b shows a single-backbone exception where KB stacks additively on SK (+0.052 via Skill-invocation re-weighting); both are single-point observations awaiting replication on non-Alibaba backbones at similar scale. A counter-result: {KB, WS} alone is net-harmful at 8 B.

- **(C4)** A cross-channel effort accounting on provider-agnostic metrics. SK raises panel-mean score while cutting panel-median tool calls by $\sim 50\%$ and output tokens by $\sim 40\%$, so the dominant score lever is also the most efficient of the three channels we test. The format-vs-task-specificity confound (SK differs from KB and WS on both axes) is addressed in §6.

Non-claims and roadmap. We do not claim a new RS-specific agent system, a new RS perception dataset, or state-of-the-art on any individual task. The rest of the paper is organised as follows. §2 surveys prior work on RS agents and knowledge augmentation; §3 formalises the design space; §4 describes the benchmark protocol; §5 presents results; §6 discusses mechanisms and a practitioner selection rule; §7 lists limitations; §8 concludes.

2 Related Work

LLM agents for remote sensing. Recent systems pair LLMs with tool harnesses to run RS analyses: RS-Agent [50], OpenEarth-Agent [59], ThinkGeo [40], Autonomous-GIS-Next-Gen [29], GeoGPT [58], LLM-Find [36], GIS-Copilot [1], OpenEarthAgent-4B-SFT [41], REMSA [10], Xiao and Ma’s change-analysis framework [47], PANGAEA-GPT [37], with a recent survey [2] cataloguing the design space.

Most systems test a single backbone and bundle their knowledge channels. The most closely related published system is RS-Agent [50], which uses a DualRAG knowledge stack on LLaMA-3.1 and Qwen-2.5 panels at 8–72 B, scored by task-planning accuracy on a tool-routing suite. We extend the line by (i) using *pixel-level scoring* (F1 for binary tasks, OA for the multi-class task) against public ground-truth rasters rather than tool-routing accuracy, (ii) factorising the bundled knowledge stack into KB×WS×SK, (iii) extending the scale span to 1.6 T MoE, and (iv) adding a compositional spatial-query task. GeoAgent [11] tests 7–8 B coding LLMs (StarCoder, LLaMA-3, DeepSeek) under fixed top-3 retrieval and reports that “RAG does not improve LLM’s generation” due to “irrelevant augmentations from suboptimal retrievers,” consistent with our 8 B KB+WS inversion (§5.4). RS vision-language work targeting text answers (GeoChat [26], RSGPT [22], EarthGPT [56]) and structured spatial-query agents (Spatial-Agent [4]) sit outside our pixel-level scoring factorial.

Benchmarks for LLM agents in RS. Two precedents anchor our methodology: AgentBench [30] for multi-environment LLM-agent evaluation, and SWE-bench [23] for execution-grounded scoring against ground truth rather than tool-call traces. SIGSPATIAL has hosted LLM-geospatial benchmarks targeting text-answer QA [5, 6], spatial-representation quality [35], and event geolocation from text [33], but none score pixel-level masks; the Sentinel-2 wildfire benchmark [31] is the closest pixel-level scoring analogue and a direct structural precedent for our T4 component. The closest RS-agent benchmarks: ThinkGeo [40] (486 tasks, step-level accuracy under a fixed knowledge bundle), GeoAnalystBench [55] (50 Python tasks, 2^2 knowledge factorial, single-model DeepSeek-R1-7B inversion), GeoBenchX [25] (LLM-as-judge), and GeoAgentBench [53] (varying agent architecture, orthogonal to our knowledge-channel axis). EarthSpatialBench [51] provides spatial-reasoning QA pairs; T5 instead measures *geometry generation*, where the agent must

emit a binary raster aligned to a 1024×1024 input grid via a multi-stage geometric pipeline with deterministic ground truth.

Skills, RAG, and scaling precedent. Our agent paradigm follows CodeAct [45] + ReAct [52] + Voyager-style skill libraries [44]. Concurrent SkillsBench [28] ablates skills against a no-skill baseline on the Claude family on terminal tasks (+14–23 pp), consistent with our main finding; we extend it to a 2^3 factorial across seven backbones and three providers in the RS domain. Balaguer et al. [3] independently document a similar contraction for the retrieval channel: the RAG benefit shrinks $\sim 6\times$ moving from a 13 B open model to GPT-4 on a domain-QA task. Flan [12] shows instruction-tuning gain shrinks $\sim 1.65\times$ as the model grows by $67\times$ in parameters; our inference-time intervention on an 8 B-to-1.6 T panel is an analogous scaling test without the weight updates standard in post-training [27].

3 Problem Formulation

Trial. A trial is a tuple $\tau = (M, T, x, K) \mapsto \hat{y}$. Here M denotes an LLM-driven agent identified by its provider and backbone model; T is a task accompanied by a natural-language prompt and a fixed input–output specification; x is a geo-referenced raster; and $K \in 2^{\{\text{KB}, \text{WS}, \text{SK}\}}$ selects which knowledge channels are exposed to the agent. The agent produces a raster mask $\hat{y}$ on x ’s grid, scored against public ground truth y^* by $\mathcal{L}(\hat{y}, y^*)$: pixel-wise F1 on the four binary tasks T1, T2, T4, T5 and overall accuracy on the multi-class task T3. The per-cell unit of analysis is the mean of $\mathcal{L}$ over trials sharing (M, T, K) . All inference is reported with paired-bootstrap 95 % confidence intervals at the scene level; the full statistical protocol is given in §5.1.

Three knowledge channels. **KB:** in-house hybrid retrieval. A curated Markdown corpus on RS fundamentals, spectral indices, raster handling, and Python libraries is partitioned into short segments. Each agent query is scored in parallel by a neural dense retriever and a lexical sparse retriever, and the two ranked lists are fused by rank-based aggregation. The top candidates are re-ranked by a cross-encoder, and segments above the relevance threshold are returned to the agent. Concrete models and indexing parameters are reported in Appendix A.1. **WS:** open-web retrieval. A live web-search tool that issues queries against the open web and returns, for each query, an ordered list of title, URL, and text-snippet entries. **SK:** procedural skill. Five task-specific procedural skills are authored as Markdown documents following the Agent Skills paradigm [49]: a short header naming the skill, followed by a body section comprising inputs–outputs contract, workflow, starting-point recipe with executable Python, scene-aware adjustments table, self-check list, and common-pitfalls section. Under progressive disclosure the agent initially sees only the headers and must invoke a skill by name to load its body; at most one skill is loaded per trial.

Open-ended vs. procedural. KB and WS are *open-ended* channels. They return reference passages on demand to supplement the agent’s own knowledge, without prescribing a course of action. SK is a *procedural* channel that hands the agent a procedure rather than asking it to author one. The three channels span the practical knowledge sources exposed to deployed RS agents, namely the

in-house knowledge base, the open web, and curated procedural artefacts.

All trials share one ReAct execution loop (Fig. 1) with a sandboxed Python executor, file primitives, and the three knowledge tools KB, WS, SK conditionally activated by K ; obfuscation, budget, and failure handling are in §4.4.

4 Experimental Setup

Imagery is Sentinel-2 across all five tasks, on a 10 m surface-reflectance grid with 20 m SWIR resampled to match. T1 and T3 share globally distributed AOIs spanning five continents (Europe, Asia, Africa, North America, Oceania); T5 spans the same five continents over its ten samples; T2 is concentrated in European and Asian urban centres through OSCD; T4 is concentrated in California and the south-western United States through CaBuAr and CHABUD.

Figure 1 summarises the evaluation pipeline. Each model-configuration-task triple defines one experimental cell, and the full 280-cell matrix scores each agent’s pixel-level output mask against a public ground-truth raster rather than against its tool-routing choices [40, 50].

4.1 Task taxonomy

The experimental suite contains five remote-sensing tasks on real public satellite imagery, summarised in Table 1. T1–T4 are established RS task families re-instrumented for pixel-level scoring; T5 is new and described separately in §4.2.

T1 Water extraction. Ten Sentinel-2 surface reflectance composites [18] on globally distributed AOIs, scored against the ESA WorldCover v200 [43, 54] water classes (codes 80/90/95); naive baseline is the MNDWI threshold [32, 48]. **T2 Bi-temporal change detection.** Ten OSCD [16] Sentinel-2 L1C pairs. T2 is the only task on L1C imagery because OSCD distributes L1C exclusively. Both epochs share L1C processing, so change-vector magnitudes are approximately preserved for OSCD’s dominant short-baseline pairs, though multi-year pairs may carry residual aerosol or BRDF offsets. The naive baseline is CVA with Otsu thresholding [14, 42]. **T3 Land cover, 4-class.** The same ten AOIs as T1 with labels derived from WorldCover by collapsing native classes into water, vegetation, built-up, and other (App. A.1); naive baseline is a published NDVI / MNDWI / NDBI rule tree [20]. The T3 procedural skill encodes the same class definitions as the ground truth, so its score partly measures recipe-following, which we treat as a known confound (§7). **T4 Burn-scar mapping.** Ten Sentinel-2 pre/post-fire pairs at 512² resolution from the CaBuAr [8] / CHABUD [9] corpus; ground-truth labels combine CAL FIRE aerial-survey delineations (CaBuAr) and Copernicus Emergency Management Service activation products (CHABUD), with per-scene provenance recorded for each sample. Naive baseline is dNBR > 0.27 [13, 24]; sweeping the cutoff over {0.10, 0.18, 0.40} yields mean F1 {0.33, 0.35, 0.30}, placing the published threshold within 0.01 of the best single-threshold value. Six of the ten scenes carry < 5 % burn coverage, so per-scene F1 is partly bounded by class imbalance (§7). **T5 Compositional spatial query.** Described in §4.2.

Pre-processing details, including the band-alignment resampling used for indices that combine 10 m and 20 m Sentinel-2 bands, are reported in App. A.1.

4.2 T5 – Compositional spatial query

T5 measures whether an agent can autonomously compose multi-class detection with a spatial join. Each sample pairs a Sentinel-2 tile with a natural-language query, for example “*Identify tree-covered areas within 300 m of any permanent water body.*” The agent must emit a binary mask that selects pixels of the queried target class whose distance to the nearest pixel of the queried reference class falls within the queried radius.

Construction. A sample (x, q, y^*) is built in four deterministic steps. We first select an ESA WorldCover v200 [54] area of interest. We then draw a target class, a reference class, and a distance r from a fixed grammar over six WorldCover land-cover classes and the discrete distance set $r \in \{150, 200, 250, 300, 400, 500\}$ m, with the spatial operator fixed to *within*. We render q through a frozen natural-language template that no model has been trained against, removing template idiosyncrasies as a confound. Finally, y^* is computed deterministically as the intersection of the target mask with the r -metre Euclidean buffer of the reference mask. All ten T5 samples carry distinct class-reference-distance tuples and distinct AOIs, removing the memorisation shortcut available to repeated-query benchmarks. The current grammar realises ten of the combinations admitted by the *within* operator; extending the operator and class sets is open future work (§7).

Engineering upper bound. Four hand-written reference pipelines parse the query and run the same four-stage flow of target detection, reference detection, distance transform, and intersection, differing only in how they acquire the class masks. Three pipelines that rely respectively on fixed thresholds, Otsu, and threshold-plus-morphological-clean-up reach F1 of 0.19, 0.19, and 0.20, defining a *spectral upper bound* of approximately 0.19; a fourth pipeline that consumes WorldCover oracle labels reaches F1 = 1.00 by construction, isolating the spectral-approximation deficit at 0.81 F1. We therefore report T5 agent scores against the absolute spectral upper bound rather than as a percentage gain. The upper-panel agent cells reach F1 = 0.22–0.24, a small but positive margin above this ceiling that should be interpreted with caution given the ten-scene sample size; the same ceiling continues to bound agent and reference pipeline alike.

Why T5 is hard. The agent must chain four stages without error, namely target detection, reference detection, buffer computation, and mask intersection. None of these is solved autonomously without procedural guidance (§5.3). The T5 procedural skill provides the four-stage scaffold with starting-point recipes for each stage; the agent must still bind the correct WorldCover class labels and chain the stages in order, so the decomposition is procedurally scaffolded rather than autonomously derived. Figure 2 shows a representative high-scoring output for each task.

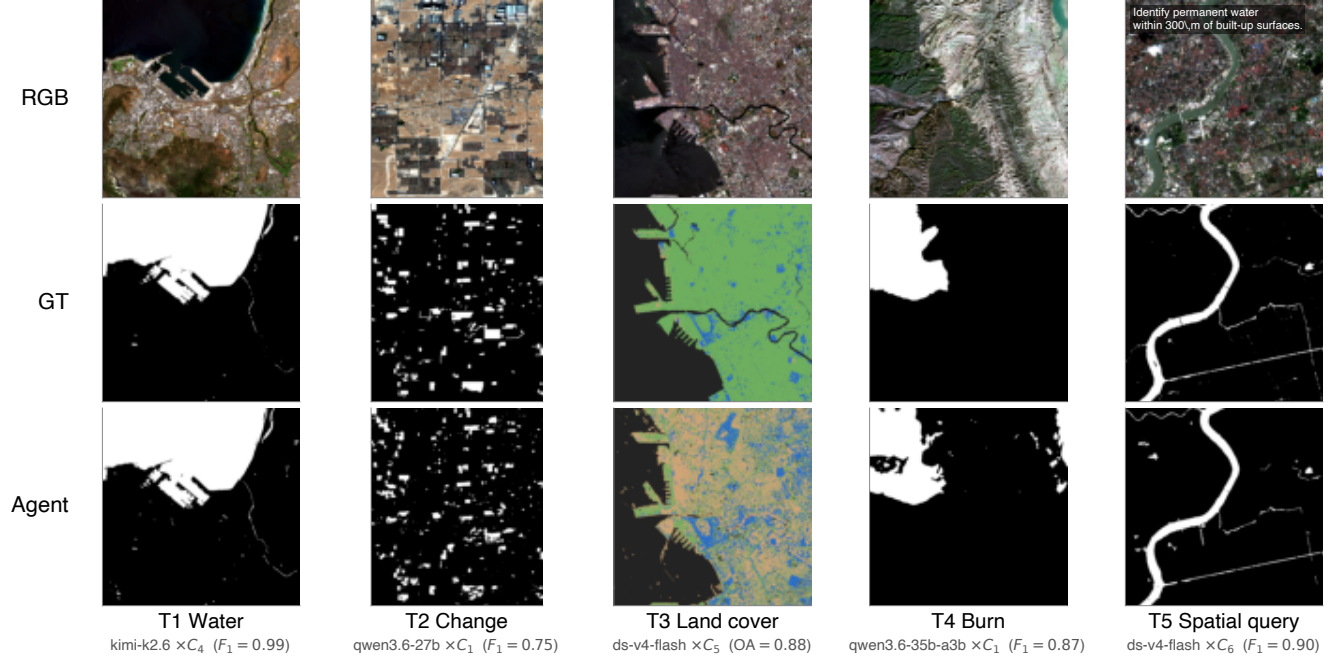
4.3 Knowledge configurations and model panel

The three knowledge channels are defined in §3. Their 2³ on-off combinations yield eight configuration cells labelled C_0 through C_7 , where the index encodes the on-off state of (KB, WS, SK) as a

Table 1: Task taxonomy. Each task supplies ten distinct scenes, replicated five times per (model, configuration) cell, giving fifty pixel-scored trials per cell. Naive baselines use published spectral-index thresholds with no task-specific tuning.

ID	Task	Imagery	Ground truth	Scenes $\times$ reps	Output mask	Metric	Naive
T1	Water extraction	Sentinel-2 SR, 10-band	WorldCover v200, cl. 80/90/95	10 $\times$ 5	binary, 1024 ²	pixel F1	0.61
T2	Bi-temporal change	OSCD Sentinel-2 L1C	OSCD labels	10 $\times$ 5	binary, variable	pixel F1	0.25
T3	Land cover, 4-class	Sentinel-2 SR, 10-band	WorldCover-derived, 4-class	10 $\times$ 5	4-class, 1024 ²	OA	0.72
T4	Burn-scar mapping	Sentinel-2 pre/post-fire	CaBuAr / CHABUD	10 $\times$ 5	binary, 512 ²	pixel F1	0.35
T5	Compositional query	Sentinel-2 SR, 10-band	WorldCover + distance buffer	10 $\times$ 5	binary, 1024 ²	pixel F1	0.12 / 0.19 [†]

[†] Naive always-NDVI baseline / hand-written reference pipeline (the spectral upper bound, §4.2).

**Figure 2: Per-task gallery.** Each column shows one high-scoring sample with model, configuration, and primary metric below. Rows are the Sentinel-2 input (rendered as false-colour RGB), the ground-truth mask, and the agent’s output mask.

three-bit vector (Figure 1). The single-channel cells C_1, C_2, C_3 isolate the marginal contribution of each channel, the two-channel cells C_4, C_5, C_6 isolate the three pairwise interactions, and the full-stack cell C_7 isolates the three-way interaction. The open-ended-only cell $C_4 = \{\text{KB}, \text{WS}\}$ tests whether retrieval alone, without procedural scaffolding, is sufficient.

The model panel spans 8 B to 1.6 T total parameters across three providers, with both dense and mixture-of-experts architectures represented (Table 2); the seven backbones are representative open-weight models. Active parameter counts are taken from each provider’s versioned model card [17, 34, 38, 39] because the regime analysis in §5.1 indexes effects on the parameter-count axis.

Closed-weight frontier models, such as the GPT, Claude, and Gemini families, cannot be placed on that axis without provider-side parameter disclosure and are deferred to future work (§7).

4.4 Trial protocol

Each task supplies ten distinct scenes, each replicated five times per (model, configuration) cell, so one cell comprises fifty trials. The

Table 2: Model panel. Active parameter count, with the total in parentheses for MoE backbones, is taken from each provider’s model card. Rows are ordered by total parameter count.

Short name	Provider	Active (total)	Arch.
qwen3-8b	Alibaba	8 B	dense
qwen3-14b	Alibaba	14 B	dense
qwen3.6-27b	Alibaba	27 B	dense
qwen3.6-35b-a3b	Alibaba	3 B (35 B)	MoE
deepseek-v4-flash	DeepSeek	13 B (284 B)	MoE
kimi-k2.6	Moonshot	32 B (1 T)	MoE
deepseek-v4-pro	DeepSeek	49 B (1.6 T)	MoE

280-cell matrix yields 14,002 scored trials (Table 4); 268 cells reach $N \geq 50$, and the remaining twelve cells on qwen3.6-35b-a3b are reported in App. A.1. Each trial proceeds in its own scratch sandbox until the agent writes its output mask or exhausts the round budget.

Per-trial obfuscation. The agent perceives only an anonymised input raster: scene identifiers, dataset names, acquisition dates, geographic coordinates, and band sidecar files are stripped before the sandbox is created, and the ground-truth raster is held in a separate directory the agent never sees. Combined with the cross-biome Sentinel-2 scene selection, this closes the shortcut by which an LLM trained on public benchmark splits could recognise the scene and recall its expected output.

Failure handling. Trials that fail to produce a valid output mask within the round budget are scored 0 on the primary metric rather than dropped, since these failures are themselves a finding (§5.5).

Statistical inference. Inferential statistics use per-cell paired-bootstrap confidence intervals over the ten scenes per task and a 2^3 ANOVA decomposition over the KB, WS, SK factors; the full resampling protocol is in §5.1.

4.5 Reproducibility

All experimental artefacts—per-trial logs, the KB corpus, the skill documents, the T5 grammar, ground-truth generation scripts, and manual failure-mode labels—are released at <https://github.com/polaris-morningstar/RS-AgentBench>, with the released per-trial scores (`all_trials.parquet`) as the canonical numerical reference for reproduction.

Use of AI tools. Implementation of the agent harness, benchmark runner, and data-processing pipelines was drafted with AI assistance (Claude, Anthropic) under author-specified design; all code was reviewed, executed, and validated by the authors.

5 Results

Across the 14,002 trials in the 280 backbone $\times$ configuration $\times$ task cells, the SK channel is the dominant lever, attenuating from Decisive +0.274 on qwen3-8b to a Saturated band of +0.001 to +0.035 across the five ≥ 27 B backbones, detailed in §5.1. Only 45 % of cells exceed a published spectral-index baseline, with the deployable margin concentrated on four of the five ≥ 27 B backbones (qwen3.6-35b-a3b excepted), with qwen3-14b a separate sub-threshold exception; see §5.2. The new compositional spatial query T5 plateaus near the spectral upper bound regardless of scale or channel, in §5.3. Task choice dominates the matrix variance at $\sigma(\text{task}) = 0.21 \gg \sigma(\text{model}) = 0.08 > \sigma(\text{cfg}) = 0.03$, and two backbones show single-point channel effects, in §5.4. SK is also the most efficient channel; cost and failure-mode accounting are in §5.5.

Submission census. 13,907 of 14,002 trials reached submission. The 95 budget-exhausted trials concentrate on qwen3-8b at 79/95, with 41 on the 8 B $\times$ C₄ $\times$ change cell alone—diagnostic of 8 B retrieval budget pressure (§5.4). $N \geq 50$ holds on 268/280 cells; bootstrap inference operates on per-task sets of $N = 10$ distinct scenes.

5.1 The SK channel as dominant lever, attenuating with backbone capability

Comparing the bare control C_0 with the structured-skill-only cell C_3 isolates the SK channel’s marginal contribution; per-model F1 means averaged across the five tasks appear in Table 3. The x-axis

uses total parameter count; active counts span 3–49 B. Under active-parameter ordering the Decisive and Saturated classifications are unchanged for the five non-MoE points, and the 3 B-active qwen3.6-35b-a3b MoE remains a dual-axis outlier regardless of ordering choice.

The seven per-model SK main-effect values, in total-parameter order, are +0.274, +0.060, +0.012, +0.035, +0.025, +0.015, +0.001; they attenuate by over two orders of magnitude, the +0.274 F1 8 B endpoint carrying a deterministic SK short-circuit (§5.4) and the +0.001 panel end lying within bootstrap noise. Each is an SK main effect (SK-on minus SK-off), distinct from the skill-only $C_3 - C_0$ column of Table 3 (App. A.5). Restricting pooling to the four perception tasks (T1, T2, T4, T5) leaves panel-pooled SK at +0.061 F1, so the T3 GT–SOP alignment does not drive the headline figure. The 8 B point is more than four times the next-largest delta of +0.060 on qwen3-14b, and the upper five points form a band centred near zero between +0.001 and +0.035. We therefore characterise regimes by KB-stack headroom, written $|\Delta_{C_5-C_3}|$ and measuring the marginal F1 gain of adding KB to an already-SK-enabled cell, rather than by the SK main effect alone.

The 1.6 T deepseek-v4-pro point (49 B active) anchors saturation: panel-pooled SK = +0.001, $|\Delta_{C_5-C_3}| = 0.003$, panel-mean F1 within a ± 0.02 band across all eight configurations, and within ± 0.04 F1 of v4-flash on every configuration (+0.003 panel-mean gap). Every lever collapses to noise here; on water the procedural channel is mildly counterproductive, each v4-pro SK-on configuration losing up to 0.04 F1 against bare.

Regime taxonomy. *Decisive*—backbone qwen3-8b, $n = 1/7$, panel-pooled SK $\geq +0.10$. SK lifts panel-mean F1 by +0.274, separating sub-baseline from above-naïve performance. The label is backbone-specific: an identical-client replication preserves the qualitative classification under server-side endpoint drift in §A.5, and the deterministic SK short-circuit of §5.4 is unique to qwen3-8b on the panel. Generalisation to non-Alibaba 8 B is open. *Transition hypothesis*—backbone qwen3-14b, $n = 1/7$, panel-pooled SK in $[+0.04, +0.10]$ and $|\Delta_{C_5-C_3}| \geq +0.04$. SK alone gives +0.060, and KB stacks additively with $\Delta_{C_5-C_3} = +0.052$ whose CI excludes zero on 2/5 tasks. The classification is pending non-Alibaba 13–20 B replication. *Saturated*—five ≥ 27 B points across three providers, dense and MoE architectures, with panel-pooled SK $\leq +0.04$ and *panel-mean* $|\Delta_{C_5-C_3}| \leq 0.010$ F1. All have Skill-invocation $\geq 98\%$ on SK-on cells, except qwen3.6-35b-a3b C_7 at 82 %; the regime is anchored by the 1.6 T deepseek-v4-pro point. The 35 B-a3b MoE is borderline-Decisive on single-cell $\Delta_{C_3-C_0} = +0.048$ but Saturated on panel-pooled SK (+0.035). Under active-parameter ordering the 13 B v4-flash sits adjacent to the 14 B Transition qwen3-14b, localising the Transition to that backbone rather than a parameter threshold (App. A.5).

Statistical robustness. Panel-pooled SK = +0.060 dominates the channel contrasts; per-model $\times$ task scene-level bootstrap CIs ($B = 2,000$) exclude zero on 14/28 perception cells uncorrected and 11/28 under Benjamini–Hochberg FDR < 0.05 (paired `sk_forest` estimator). Full contrasts, the 2^3 ANOVA decomposition, and the per-(model, task) SK table (Table 7) are in Appendix A.5.

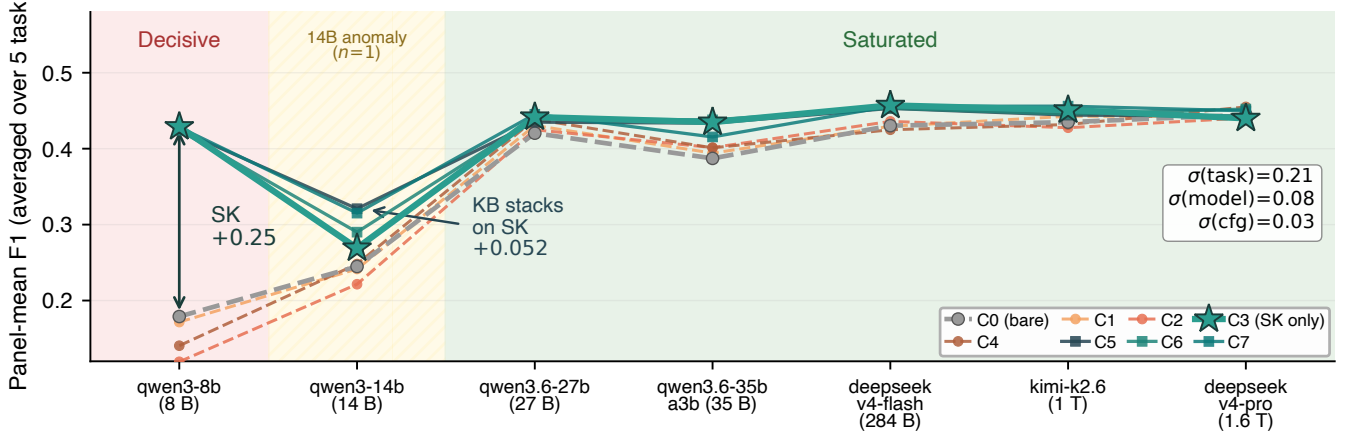


Figure 3: Per-model panel-mean F1 across the seven backbones (ordinal x-axis) for all eight configurations. Bands mark the Decisive observation and the Saturated regime; the middle band is a single-backbone anomaly (qwen3-14b, $n=1$), not a regime.

Table 3: Headline configuration grid: per-(model, configuration) panel-mean of the five task scores. Primary metric is pixel F1 on T1, T2, T4, T5 and overall accuracy on T3; T3 OA carries a GT-SOP alignment by construction (the procedural skill encodes WorldCover-consistent class thresholds; see §4.1 and §7), so per-cell SK lifts on T3 should be read as partially recipe-following rather than pure classification capability. Bold marks each row’s best cell. [‡] qwen3-8b SK-on cells emit identical masks across reps—a deterministic SK short-circuit with effective $N=10$ scenes; an identical-client replication preserved the Decisive classification under server-side drift.

Model	Active	C_0	C_1	C_2	C_3	C_4	C_5	C_6	C_7
qwen3-8b [‡]	8 B	0.179	0.172	0.119	0.429	0.141	0.425	0.424	0.428
qwen3-14b	14 B	0.245	0.242	0.221	0.269	0.248	0.321	0.290	0.315
qwen3.6-27b	27 B	0.420	0.431	0.425	0.442	0.439	0.435	0.439	0.445
qwen3.6-35b-a3b	3 B (35 B)	0.387	0.394	0.401	0.435	0.401	0.434	0.438	0.415
deepseek-v4-flash	13 B (284 B)	0.430	0.429	0.436	0.457	0.425	0.453	0.453	0.456
kimi-k2.6	32 B (1 T)	0.435	0.443	0.428	0.451	0.432	0.444	0.446	0.456
deepseek-v4-pro	49 B (1.6 T)	0.445	0.443	0.440	0.440	0.455	0.443	0.452	0.450

5.2 Where agents exceed the spectral baseline

Panel-mean F1 alone does not say whether an agent meaningfully beats a much simpler procedure: a published spectral-index threshold. We compare each cell’s mean primary metric against the per-task naive baseline defined in §4.1: F1 of 0.61 on T1, 0.25 on T2, 0.35 on T4, 0.19 on T5; OA of 0.72 on T3 (the T5 0.19 is its spectral upper bound, not a naive baseline). Across the 280 cells, 125 (44.6 %) are at or above the baseline. Figure 4 resolves the panel into per-cell deployable margins split by the SK channel, so the backbone and channel effects on deployability are visible in a single view.

Backbone axis: scale matters with two exceptions. Four of the five ≥ 27 B backbones exceed the baseline on $\geq 25/40$ cells: deepseek-v4-pro leads at 30/40 (75 %), with qwen3.6-27b, deepseek-v4-flash, and kimi-k2.6 at 25–26/40 (62–65 %). The two exceptions have lower effective capacity than their totals suggest: qwen3.6-35b-a3b (≥ 27 B, 3 B active) at 9/40, and the sub-threshold qwen3-14b at 1/40. The qwen3-14b case is the more notable. Across all five tasks and all eight configurations it produces only one deployable cell, change $\times C_5$, despite a panel-mean F1 of 0.269 close to qwen3-8b’s 0.290. Under active-parameter ordering, qwen3.6-35b-a3b’s 3 B slot

is consistent with its weak deployable performance; qwen3-14b remains the genuine deployment outlier.

Configuration axis: SK separates above- from below-naive. The four SK-on configurations C_3, C_5, C_6, C_7 each clear the spectral baseline on 18–20 of the 35 model $\times$ task combinations (51–57 %). The four SK-off configurations C_0, C_1, C_2, C_4 cluster at 11–14/35 (31–40 %). The two weakest channels are the WS-only C_2 and the KB+WS combination C_4 , both at 11/35 (31 %). Open-ended retrieval without procedural scaffolding therefore produces fewer deployable cells than the bare control C_0 on this axis. SK is the dominant lever (§5.1) not just in panel-mean F1 but in whether any given model $\times$ task cell becomes deployable at all.

Task axis: water and change deploy, burn does not. The five tasks divide into three regimes by deployability. T1 water and T2 change are the most tractable, with 37/56 cells (66 %) above naive on each, and the four ≥ 27 B backbones with ≥ 13 B active parameters cover $\geq 7/8$ configurations on water and $\geq 5/8$ on change. T5 query occupies the middle ground at 30/56 (54 %), bounded by the spectral upper bound discussed next. T3 landcover and T4 burn are the deployment floor: 13/56 (23 %) on landcover and only 8/56 (14 %)

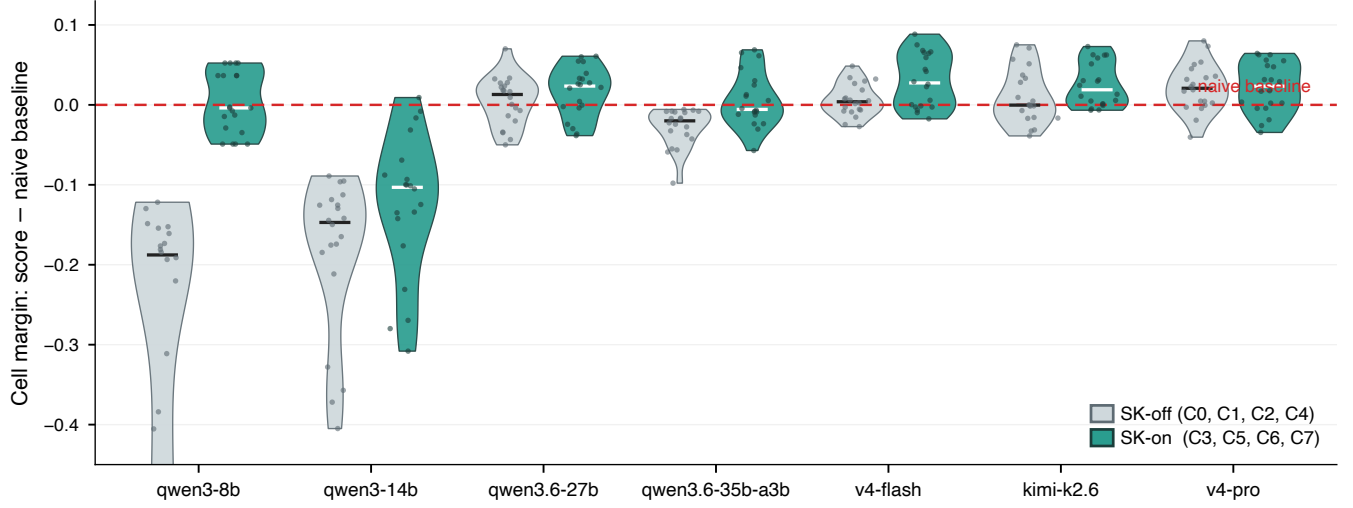


Figure 4: Per-cell deployable margin (agent score minus the task’s naive spectral-index baseline). Each model column holds two violins: SK-off cells (C_0, C_1, C_2, C_4 ; $n=20$ each, grey) and SK-on cells (C_3, C_5, C_6, C_7 ; $n=20$ each, teal). White / dark bars mark medians; inner dots show individual cells. The red dashed line at zero is the deployment threshold.

on burn. The burn figure makes T4 the one task where no backbone exceeds the naive baseline on a majority of configurations. The naive baselines themselves explain the gradient: $\text{dNBR} > 0.27$ and the $\text{NDVI}/\text{MNDWI}/\text{NDBI}$ tree-rule are themselves strong heuristics on burn and landcover, so the agent has little headroom to exploit.

5.3 T5 plateaus at the spectral upper bound

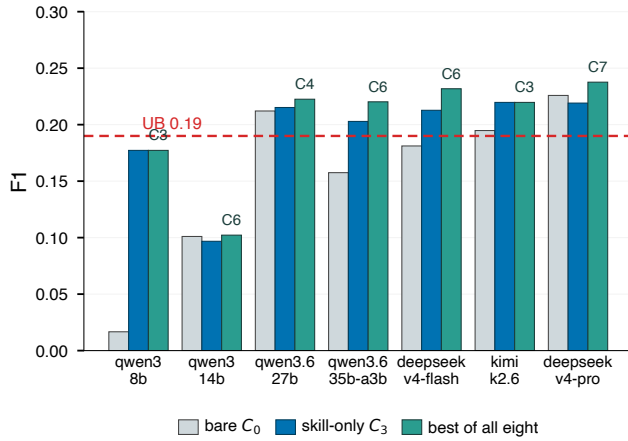


Figure 5: T5 per-model F1: bare C_0 (grey), skill-only C_3 (blue), and best of all eight configurations (teal, with the winning configuration annotated above each teal bar). Dashed line marks the spectral upper bound at 0.19.

T5 requires compositional spatial reasoning: given a target class, a reference class, and a buffer radius r , the agent must return all target-class pixels within r metres of any reference-class object. The task can typically be addressed by a four-step pipeline—target

detection, reference detection, distance-buffer construction, mask intersection—and we generate ground truth by running this same pipeline over an oracle WorldCover label source. The spectral upper bound $\text{UB} = 0.19$ F1 therefore reflects the class-grounding ceiling alone; the spatial operator contributes negligibly.

Three observations are visible in Figure 5. First, the 8 B model cannot produce useful T5 masks without a structured SOP: its bare F1 is 0.017 and SK lifts it to 0.177, a +0.16 single-cell jump. Second, qwen3-14b is the one panel point that does not cross UB, with best cell $C_6 = 0.102$; every ≥ 27 B backbone reaches or modestly exceeds it, with the deepseek-v4-pro $\times C_7$ cell topping the panel at 0.238. Third, no configuration at any scale meaningfully exceeds the spectral upper bound: the five upper-panel best cells score 0.22–0.24 F1, a 0.03–0.05 F1 margin above UB but still far below the oracle pipeline’s 1.000, and the margin should be interpreted with caution given $n = 10$ samples per task. deepseek-v4-pro reaches $\text{F1} = 0.226$ at the bare C_0 cell already, indicating that its T5 ceiling is fixed by spectral class-grounding ability rather than by access to a procedural SOP.

On T5, SK is *necessary but not sufficient*: it lifts the small-model floor to near-UB performance, but no agent meaningfully exceeds UB. T5 also carries the highest full-failure ($\text{F1} = 0$) rate of any task at a pooled 18 % (9–44 % across backbones) against ≤ 3 % on T2/T3; per-scene F1 on deepseek-v4-flash $\times C_3$ spans 0.000–0.731, so the 0.22–0.24 best-cell figure is a panel-mean over a bimodal per-scene distribution.

Manual inspection of deepseek-v4-flash’s C_7 outputs shows that residual failures are dominated by upstream class-grounding—mapping “built-up area” or “permanent water body” to a spectral threshold—not by the spatial operator itself; we return to this in §6.

5.4 Variance decomposition and backbone-specific effects

Where the variance concentrates. Task choice carries roughly $2.8\times$ the variance of model scale and $6.6\times$ that of channel configuration. Writing $\sigma(\text{factor})$ for the sample standard deviation of the per-cell mean primary metrics grouped by that factor, $\sigma(\text{task})=0.215$ across per-task grand means of burn 0.30, change 0.25, landcover 0.68, query 0.17, water 0.55 (T3 reports OA and the other four tasks report F1; F1-only σ across T1/T2/T4/T5 is ≈ 0.16 and still dominates the other two factors); the corresponding $\sigma(\text{model}) = 0.077$ and $\sigma(\text{cfg}) = 0.032$. The deployment implication is direct: choosing which RS task to automate carries more risk than choosing which backbone or which knowledge configuration to use. The SK lever is also task-dependent. The panel-pooled SK main effect is $+0.10$ F1 on water, $+0.08$ on change, $+0.06$ on landcover, $+0.03$ on burn, and $+0.03$ on T5—roughly $3\times$ stronger on water than on T5. T3 OA carries an additional caveat: WorldCover’s $\sim 76\%$ label accuracy and the proximity of the SK SOP’s class definitions to the GT label source, discussed in §4.1, make T3 difficult to interpret as pure agent capability.

The 14B exception: KB additive on SK via Skill-invocation re-weighting. qwen3-14b’s Skill-invocation rate at C_3 is only 14.8%, against $\geq 98.4\%$ on every other backbone; adding KB raises it to 57.6% on C_5 (WS alone only to 28.8%), so KB drives the climb. The $\Delta_{C_5-C_3} = +0.052$ F1 lift is mostly re-weighting trials into the higher-yielding Skill-invoked branch (F1 0.37–0.54 vs. bypass 0.16–0.25) rather than KB content. With $n = 1$ this characterises one backbone’s low-invocation tendency, not a generalisable regime; the per-branch decomposition and replication checks are in Appendix A.4.

8B-specific patterns: SK short-circuit and KB+WS inversion. qwen3-8b’s SK-on cells emit pixel-identical predicted masks across repetitions, so the $8B \times SK$ F1 should not be extrapolated outside the panel distribution. The open-ended pair $C_4 = \{KB, WS\}$ inverts 8B’s T1–T4 mean from 0.220 to 0.159 as retrieve loops exhaust the budget, while $\geq 27B$ KB/WS main effects lie within ± 0.01 F1 of zero—consistent with the GeoAnalystBench DeepSeek-R1-7B finding [55].

5.5 Effort accounting and failure modes

Effort distribution. Enabling SK over C_0 raises panel-mean F1 by $+0.054$ and cuts panel-median tool-call counts from 13 to 8, a $\sim 38\%$ reduction on the direct $C_0 \rightarrow C_3$ comparison or $\sim 50\%$ when SK-off and SK-on configuration pools are compared. Output-token estimates (elapsed wall-clock $\times$ a backbone-specific output-rate constant) drop $\sim 40\%$ over the same comparison; cross-provider absolutes are wall-clock-derived proxies, while same-backbone SK contrasts are unaffected. Figure 6 renders the resulting quality-effort separation. The open-ended channels C_1, C_2, C_4 consume similar or higher tool-call counts for $\Delta F1$ within ± 0.013 . Per-trial C_3 output tokens span $\sim 14.2\times$ across backbones ($\sim 3.1\times$ excluding the qwen3-8b deterministic outlier, §5.4), so absolute token cost is a backbone- not channel-choice question.

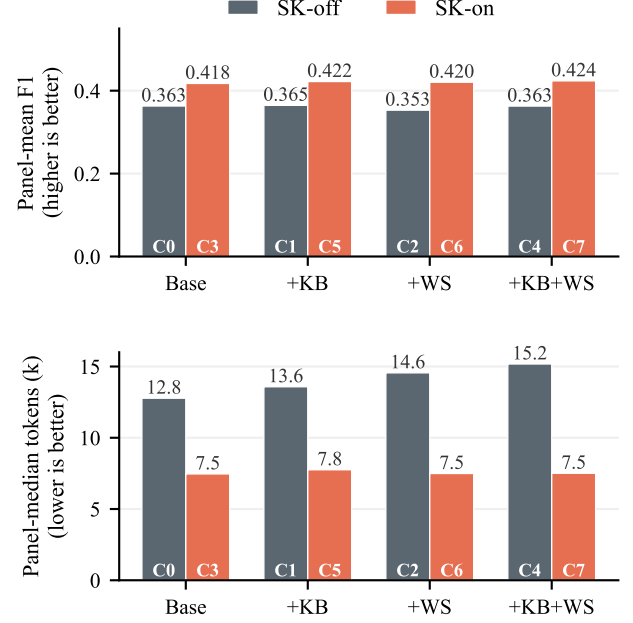


Figure 6: SK lift on quality (top) and cost (bottom) across the four cfg pairings Base / +KB / +WS / +KB+WS. Each pair differs only by the SK toggle, so the same-pair comparison isolates the SK contribution. SK-on raises panel-mean F1 by about 0.06 and cuts panel-median output tokens by about 5–8 K in every pair; the same direction holds on panel-median tool calls, reported in the text.

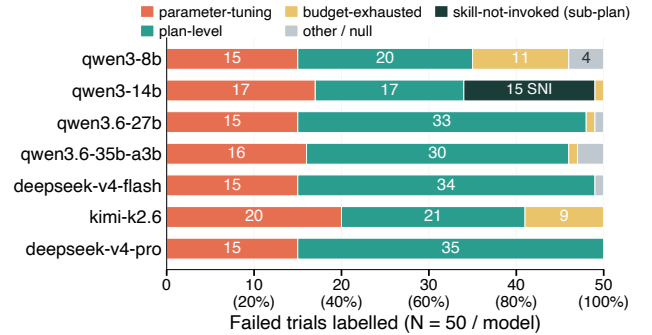


Figure 7: Manual failure-mode category distribution per model ($N=350$ stratified, 50 per model across all seven backbones).

Failure-mode taxonomy. We manually label $N = 350$ failed trials, 50 per backbone, under an 11-category schema; the rubric (Table 5) is in Appendix A.2. Collapsing the schema into four super-categories yields the four-category distribution shown in Figure 7. *Parameter-tuning* errors—wrong band index or wrong index-threshold cutoff inside otherwise runnable code—occupy 30–40% of every backbone with no scale gradient, peaking at 20/50 on

kimi-k2.6, all of them threshold-choice. *Plan-level* errors comprise class-confusion, compositional, and skill-not-invoked failures, and dominate on five of the seven backbones at 60–70 %. deepseek-v4-pro peaks at 35/50, of which 26 are class-confusion (12 on OSCD semantic-vs-spectral change). The two exceptions are qwen3-8b at 40 % plan-level and kimi-k2.6 at 42 %; both divert the remaining share into *budget-exhausted stalls*—retrieve loops that never reach a code commit, at 22 % on qwen3-8b and 18 % on kimi-k2.6, against ≤ 2 % on the other five and zero on the 1.6 T v4-pro. qwen3-14b carries a unique skill-not-invoked streak of 15/50 against zero elsewhere, consistent with its 40 % pooled Skill-invocation across SK-on cells (§5.4). *True code-execution errors*—hallucinated function, malformed output, wrong input file—are negligible at 4/350, or 1.1 %. Pooled, parameter-tuning (32 %) and plan-level (59 %) errors account for 91 % of the labelled pool (95 % binomial (Clopper–Pearson) CI [87, 94]). Agents reliably produce runnable code; the dominant failure surface is RS-specific decision-making—which band to use, which threshold to set, which class to bind—rather than coding ability. The `crs_ignored` class received zero labels across the entire 350-trial pool, indicating that current backbones already handle CRS bookkeeping reliably on the panel’s optical-only inputs.

6 Discussion

Channels: SK is the dominant lever in both score and effort. Of the three channels only SK moves panel-mean F1, and it does so while reducing effort rather than adding it—the pooled SK main effect is +0.060 F1 (direct $C_0 \rightarrow C_3$ +0.054) alongside fewer tool calls ($13 \rightarrow 8$) and ~40 % fewer output tokens (§5.5). KB and WS main effects stay within ± 0.01 F1 of zero on every ≥ 27 B backbone, and their open-ended combination is net-harmful on 8 B (§5.4): supplying the same RS content as retrieved chunks rather than as procedure does not substitute for SK.

Scale: the SK lever attenuates to a Saturated regime above 27 B. The SK main effect attenuates by more than two orders of magnitude across the panel. qwen3-8b is Decisive at +0.274 F1, but its SK-on outputs are pixel-identical across repetitions—a deterministic SK short-circuit rather than stochastic generalisation (§5.4). qwen3-14b alone shows the Transition anomaly (panel-pooled SK +0.060): the only panel point where KB stacks additively on SK, at $\Delta_{C_5-C_3} = +0.052$ via Skill-invocation re-weighting ($n = 1$ of 7). The remaining five backbones (three providers, dense and MoE, ~60× parameter range) all lie within *panel-mean* $|\Delta_{C_5-C_3}| \leq 0.010$ F1—the Saturated regime. The contraction is sharper than analogous scaling for retrieval [3] or instruction-tuning [12], consistent with SK encoding RS-specific calibration directly.

The bottleneck is RS-specific decision-making, not coding ability. The failure taxonomy localises the gap: agents reliably produce runnable code—code-execution errors are 1.1 % of 350 labelled failures—but fail at the RS-specific choice of band, threshold, class, and compositional plan that the parameter-tuning and plan-level categories (91 % combined) capture (§5.5). SK encodes exactly this content as procedure. Future RS-agent improvement should target richer procedural SOPs and class-grounding tools, not larger backbones.

A practitioner selection rule (hypothesis from $n = 1$). The KB-on-top-of-SK trigger rule rests on the single qwen3-14b observation. For deployments on this panel: probe Skill-invocation rate on ~50 held-out SK-on trials; if < 90 % enable KB on top of SK, otherwise the KB lever yields ≤ 0.010 F1. The coarser “ ≥ 27 B-total $\Rightarrow$ Saturated” heuristic agrees on 6/7 points. WS adds $|\Delta_{C_6-C_3}| \leq 0.006$ F1 on five of six non-14 B backbones (deepseek-v4-pro excepted at +0.012), so disabling WS by default is defensible.

Format vs. specificity confound. SK differs from KB and WS on both format and task specificity; isolating which drives the gain via a controlled task-specific KB is open future work.

7 Limitations

Statistical power. Each task uses $N = 10$ scenes, each replicated five times. SK main-effect bootstrap CIs exclude zero on 11/28 perception cells under Benjamini–Hochberg FDR < 0.05 (§5.1). The Decisive observation and Transition hypothesis are $n = 1$ of 7 and await replication.

Panel and task scope. Four of seven backbones are from Alibaba; the Saturated regime has non-Alibaba support, but the Transition hypothesis awaits non-Alibaba 13–20 B replication. Closed-weight frontier models—GPT, Claude, Gemini—cannot be placed on the active/total parameter axis. Tasks are optical-only; T5 exercises the *within-operator* on 10 of 180 in-grammar tuples.

Dataset, labelling, and effort metrics. T2 OSCD is European/Asian urban change; T4 CaBuAr/CHABUD is predominantly western-US Mediterranean burns, and six of the ten T4 scenes have < 5 % burn-pixel coverage so per-scene F1 is structurally bounded by class imbalance; T3 uses WorldCover v200 as both input and label, capping OA near its ~76 % class-averaged label-noise floor [54]. The 350-trial failure taxonomy (§5.5) sub-samples 14,002 trials under fixed decision-order labelling, with inter-rater $\kappa = 0.51$ on the four super-categories (App. A.2).

8 Conclusion

We presented RS-AgentBench, a pixel-level benchmark factorising three knowledge channels across an open-weight scaling panel and five remote-sensing tasks. We find a Saturated regime above 27 B in which channel choice is largely interchangeable on score; the single dominant lever, task-specific procedural skills, raises score and reduces effort together. Failure-mode analysis points to RS-specific decision-making—not coding ability—as the next bottleneck. All artefacts are released for replication.

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A Appendix

A.1 Reproducibility configuration

All seven backbones run at temperature = 0.6 with seed=42 and extended chain-of-thought thinking enabled where supported. Inference uses $B=2,000$ scene-level paired-bootstrap confidence intervals with NumPy seed 42. KB is indexed by a hybrid pipeline that combines a neural dense embedder [57] and a BM25 sparse retriever fused by Reciprocal Rank Fusion [15] with $k=60$ to top-10 candidates, re-ranked by a cross-encoder and returned as the top-5 above a relevance threshold of 0.30. WS uses the Tavily search service. SK is a library of five Markdown skill documents following the template in §3. For spectral indices that combine 10 m visible/NIR bands with 20 m SWIR, the SWIR band is bilinearly resampled to the 10 m grid before per-pixel index computation. The T3 four-class ground truth collapses the eleven WorldCover v200 classes as water = {80, 90, 95} (permanent water, herbaceous wetland, mangrove), vegetation = {10, 20, 30, 40} (tree, shrub, grass, crop), built-up = {50},

and other = {60, 70, 100} (bare/sparse, snow/ice, moss/lichen); class 0 (no-data) is ignored.

Table 4: Cell-count audit. The 280-cell matrix yields 14,002 scored trials. qwen3.6-35b-a3b has 12 cells short of $N=50$; 10 stay above $N \geq 43$ (≤ 0.01 F1 vs. full- N re-runs), and the two smallest ($C_{\text{water}} N=20$, query $N=34$) do not affect the Saturated classification.

Model	Trials	$N \geq 50$	$N=20-49$
qwen3-8b	2,064	40	0
qwen3-14b	2,000	40	0
qwen3.6-27b	2,000	40	0
qwen3.6-35b-a3b	1,938	28	12
deepseek-v4-flash	2,000	40	0
deepseek-v4-pro	2,000	40	0
kimi-k2.6	2,000	40	0
Total	14,002	268	12

A.2 Failure-mode schema and labels

Pool B contains 350 failed trials, 50 per backbone, sampled stratified from cells with primary metric < 0.10 or unsubmitted. Each trial received a single primary label drawn from the 11-category schema by manual review under a fixed decision order; the full rubric and per-trial labels are released for reproduction. To estimate inter-rater reliability we re-labelled a stratified $N=50$ subsample blind under the same rubric: observed agreement was 30/50 (0.60), Cohen’s $\kappa=0.49$ on the 11-class taxonomy and $\kappa=0.51$ on the four super-categories of Figure 7; the principal confusion is class_confusion $\leftrightarrow$ ndvi_threshold. The four super-categories used in the body text (parameter-tuning, plan-level, budget-exhausted, other/null) collapse the 11 classes; counts in this appendix are over the 350-trial pool.

Table 5: The 11-category manual failure schema and counts ($N=350$). Decision order is top-to-bottom: when multiple classes apply, the first wins.

Class	Definition	N
band_index	Wrong Sentinel-2 band index	13
ndvi_threshold	Correct index, wrong threshold	100
crs_ignored	CRS or projection ignored	0
wrong_input	Read wrong file	0
skill_not_invoked	SK cfg but no Skill() call	15
hallucinated_fn	Called non-existent function	2
timeout_stall	Tool-call budget exhausted	23
malformed_output	Wrong shape, dtype, or value range	2
class_confusion	Wrong semantic class	116
compositional_plan	Wrong T5 (<i>target</i> , <i>ref</i> , <i>r</i>)	74
other	Unclassifiable / multi-cause	5

A.3 T5 spatial-query grammar

All ten T5 samples use the *within*-operator on AOIs spanning five continents (Europe, Asia, Africa, North America, Oceania).

Six WorldCover v200 classes appear as targets or references, paired with six buffer radii {150, 200, 250, 300, 400, 500} m. All ten (*target, reference, r*) tuples are distinct; five of the ten use permanent water as the reference class (the most spectrally separable WorldCover class), so the spectral UB = 0.19 may be optimistic for non-water reference classes. Per-sample names and tuples are released in `data/t5_query_tuples.json`; Figure 8 illustrates a representative class-grounding plus buffer-omission failure on one tuple.

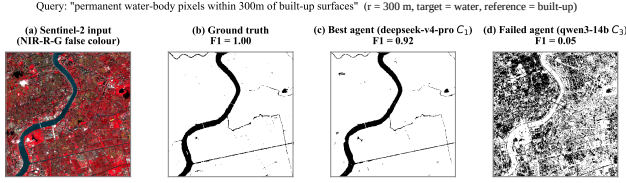


Figure 8: T5 spatial failure illustration on yangtze_delta_cn_t5. Query: “water-body pixels within 300 m of built-up surfaces.” (a) Sentinel-2 NIR/R/G false-colour input. (b) Ground truth at F1 = 1.00. (c) deepseek-v4-pro $\times C_1$ matches the buffered-water geometry at F1 = 0.92. (d) qwen3-14b $\times C_3$ over-predicts built-up coverage and skips the buffer step, producing F1 = 0.05 with 39 % positive coverage against the GT’s 7 %—a class-grounding plus buffer-omission failure mode.

A.4 The 14 B Skill-invocation anomaly

Six of seven backbones invoke Skill on $\geq 98\%$ of SK-on trials (Table 6); qwen3-14b is the only backbone with a configuration-sensitive rate (KB raises it from 14.8 % at C_3 to 57.6 % at C_5 , WS only to 28.8 %), and the lone other sub-98 % cell is qwen3.6-35b-a3b $\times C_7$ at 82.4 %. For the $\Delta_{C_5-C_3} = +0.052$ lift the Skill-invoked branch yields F1 = 0.442 at C_3 against bypass 0.239; the lift is +0.087 from re-weighting trials into that higher-yielding branch, offset by a -0.035 within-branch KB-induced loss. An identical-client replication reproduces the pattern within 1–3 pp; the parallel 8 B replication drifted on SK-on cells, ruling out environmental causes.

Table 6: Skill-invocation rate (%) per (model, configuration) on the four SK-on cells. qwen3-14b is the sole backbone with a configuration-sensitive rate; the lone other sub-98 % cell is qwen3.6-35b-a3b $\times C_7$ at 82.4 %.

Model	C_3	C_5	C_6	C_7
qwen3-8b	99.6	99.2	100.0	100.0
qwen3-14b	14.8	57.6	28.8	59.2
qwen3.6-27b	100.0	100.0	100.0	100.0
qwen3.6-35b-a3b	100.0	100.0	100.0	82.4
deepseek-v4-flash	100.0	100.0	100.0	100.0
deepseek-v4-pro	100.0	100.0	100.0	100.0
kimi-k2.6	100.0	98.4	100.0	99.2

A.5 Statistical detail and active-parameter robustness

Panel-pooled contrasts on F1: SK = +0.060, KB = +0.005, WS = -0.002 , all interactions ≤ 0.005 F1 at cell-mean aggregation. The per-model SK values in §5.1 are SK *main effects*: per-scene mean over SK-on (C_3, C_5, C_6, C_7) minus SK-off (C_0, C_1, C_2, C_4), averaged across tasks (paired sk_forest estimator; per-(model, task) terms in Table 7). This differs from the skill-only $C_3 - C_0$ column of Table 3: the SK main effect is +0.274 vs $C_3 - C_0$ +0.250 at qwen3-8b, +0.060 vs +0.024 at qwen3-14b. Per-model $\times$ task SK bootstrap CIs ($B=2,000$, scene-level) exclude zero on 14/28 perception cells uncorrected and 11/28 under Benjamini–Hochberg FDR < 0.05 (full table released); the qwen3-8b 4/4 tally is determinism-derived (§5.4). The 2^3 ANOVA shows SK dominating 8 B variance; on the five Saturated rows, channel factors explain $\leq 3.1\%$ of variance, no off-SK term reaching 1 % SS—the sole exception above 8 B being 14 B $\times$ change (KB $F=22.8$, $p < 0.001$, 4.9 % SS).

Table 7: Per-(model, task) SK main effect (mean F1 SK-on minus SK-off; T3 reports OA; tasks T1–T5 as in Table 1), in total-parameter order. Mean is the per-model SK effect of §5.1; dropping the GT-SOP-aligned T3 column leaves panel-pooled SK at +0.061 F1, with per-cell 95 % CIs released.

Model	T1	T2	T3	T4	T5	Mean
qwen3-8b	+0.547	+0.230	+0.308	+0.161	+0.123	+0.274
qwen3-14b	+0.093	+0.101	+0.072	+0.037	−0.004	+0.060
qwen3.6-27b	−0.007	+0.049	+0.012	+0.002	+0.003	+0.012
qwen3.6-35b-a3b	+0.047	+0.074	+0.005	+0.018	+0.034	+0.035
deepseek-v4-flash	+0.031	+0.077	−0.007	−0.003	+0.026	+0.025
kimi-k2.6	+0.015	+0.001	+0.020	+0.022	+0.017	+0.015
deepseek-v4-pro	−0.012	+0.021	−0.002	−0.004	0.000	+0.001

Active-parameter robustness. Under active-parameter ordering (3, 8, 13, 14, 27, 32, 49 B), per-model $\Delta_{C_3-C_0}$ values are +0.048, +0.250, +0.027, +0.024, +0.021, +0.016, -0.005 . The Decisive and Transition labels survive the reorder (the Saturated 13 B v4-flash sits adjacent to the Transition 14 B point), and the dense-equivalent conversion $\sqrt{N_{\text{active}} \cdot N_{\text{total}}}$ [19] preserves Saturated under all three orderings. A bit-identical replication of qwen3-8b reproduces *Decisive*; 14 B showed no drift.